QUIZ ONE Prof. Magda Haggag

Status	Finished
Started	Friday, 27 March 2026, 1:41 PM
Completed	Friday, 27 March 2026, 1:43 PM
Duration	1 min 42 secs
Grade	**8.00** out of 10.00 **(80%)**

Question **1**

Correct

Mark 1.00 out of 1.00

Flag question

Let X1, X2, ...,, X10 be 10 independent and identically distributed (iid) random variables taking the values 0 , 1 with corresponding probabilities p, q. Then X = X1+ X2 +... + X10 is / has:

a. A normal distribution

b. Poisson distribution

c. A binomial distribution ✓

d. A geometric Distribution

Your answer is correct.

The correct answer is: A binomial distribution

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Question **2**

Correct

Mark 1.00 out of 1.00

Flag question

Let X follows Poisson distribution with mean λ .. If P(X=2) = 0.5P(X=3), then the value of λ is

a. 4

b. 2

c. 8

d. 6

Your answer is correct.

The correct answer is: 6

Question **3**

Incorrect

Mark 0.00 out of 1.00

Flag question

Let X and be two independent Poisson variables with rate 5 and 10, respectively , then E(XY) is equal to:

a. 2

b. 5

c. 0.5

d. 10

Your answer is incorrect.

The correct answer is: 5

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Question **4**

Correct

Mark 1.00 out of 1.00

Flag question

If P(1) = P(5) in Poisson's Distribution, find the value of the mean

- a. 1.33
- b. 3.38
- c. 3.31 ✓
- d. 5.38

Your answer is correct.

The correct answer is: 3.31

Question **5**

Correct

Mark 1.00 out of 1.00

Flag question

Which of the following is the standard deviation of binomial distribution?

- a. np
- b. 2npq
- c. npq
- d. (npq)^0.5 ✓

Your answer is correct.

The correct answer is: (npq)^0.5

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Let X e a discrete random variable which has a probability mass function $f(x) = c\,x$ for $x=1,2,3,4$ and zero otherwise. hat is the constant c?

- a. 0.01
- b. 0.005
- c. 0.1 ✓
- d. 0.5

Your answer is correct.

The correct answer is: 0.1

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If X is a discrete r.vv. such that:

X	f(x)
3	0.4
4	0.3
5	0.3

Find E(x)

- a. 2.7
- b. 5.2
- c. 3.9 ✓
- d. 7.9

Your answer is correct.

The correct answer is: 3.9

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Question **8**

Correct

Mark 1.00 out of 1.00

Flag question

The number of customers arriving at a grocery store is a Poisson r.v. . On average 10 customers arrive per hour. Let X be the number of customers arriving from 10am to 11.30am . What is the probability that:

$P(10 < X \leq 15)$?

- a. 0.2269
- b. 0.4496 ✓
- c. 0.3964
- d. 0.6644

Your answer is correct.

The correct answer is: 0.4496

Question **9**

Correct

Mark 1.00 out of 1.00

Flag question

A probability distribution showing the probability of x successes in n trials, where the probability of success does not change from trial to trial, is termed a

- a. geometric probability distribution
- b. normal probability distribution
- c. binomial probability distribution ✓
- d. uniform probability distribution

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A description of the distribution of the values of a random variable and their associated probabilities is called a

a. probability distribution

b. random variable

c. expected value

d. random variane ⊗

Your answer is incorrect.

The correct answer is: probability distribution

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